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Github link : https://github.com/chavaanilkumar/traffic-congestion-prediction-and-optimisation-using-real-time-urban-mobility-and-sensor-data/blob/main/Code.ipynb

Machine learning–based intelligent traffic congestion prediction and optimisation using real-time urban mobility and sensor data

Abstract

This research project is aimed at creating a predictive model of the speed of traffic and detecting congestion with the help of machine learning methods. The issue of traffic jams in urban areas is very problematic, as it brings about more time to travel, fuel consumption, and exhaustion on the environment. To alleviate this, the study employs the real-time traffic information of the vehicle speed, flow, occupancy, and temporal characteristics of traffic data of urban sensor networks. There were several machine learning models implemented and tested, such as Linear Regression, Support Vector Regression, Random Forest Regressor, and Gradient Boosting Regressor. Gradient boosting was also the most accurate in its prediction, which is able to capture non-linear traffic trends. Modeling models that used feature engineering did not respond to systemic reliability but rather focused on time evidence speed, lagged speeds, and time indicators. Congestion detection using threshold-based classification was also included in the study with great reliability in the differentiation of free flow and congested traffic conditions. The results confirm the significance of a data-driven approach in the context of intelligent transportation systems and illustrate how predictive models can be used to facilitate real-time traffic management and decision-making.

Acknowledgement

My supervisor will be the person to whom I would like to say that my gratitude for this project was remarkable. It is also because my institution has provided the required resources and learning environment. Lastly, I would like to express my gratitude to my family and friends that encouraged and motivated me, which led to the successful completion of the present research work.

Declaration

I am, therefore, claiming that the presented project is my original work and has never been presented earlier to acquire any academic award. All the information sources have been properly acknowledged and cited. I can affirm that the present work was done during the adherence to the principles of academic integrity and is not related to any cases of plagiarism or unauthorized help.

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Chapter 1: Introduction

1.1 Overview of the Project

Traffic jams in urban settings are a burning issue in modern society as a result of both intensive urbanization and increased car ownership coupled with the intricacy of the road networks, which causes bottlenecks. The effects of these congestion events are the increased time to travel, consumption of more fuel, and levels of environmental pollutants, which directly affect the sustainability of urban environments and satisfaction of the commuters. The conventional traffic management tools, which are generally dependent on fixed signal timings and trial and error adjustments, are not sufficient to accommodate traffic dynamics. Due to these reasons, the need to develop intelligent and adaptive solutions that will take advantage of recent data-driven methods increases (Cirianni et al., 2023).

The project is aimed at using machine learning models to forecast and optimize real-time congestion. Through access to the urban mobility datasets and GPS traces and sensor networks, the study will come up with predictive models that precisely predict the speed of traffic and congestion events. The developed impact of these models is supposed to guide the traffic control models, such as adjustable signal time and optimized route plans, that can lead to boosting the efficiency and dependability of intelligent transportation systems (ITS). This is a method of filling the gap between theoretical predictive models and pragmatic urban traffic management such that practical actionable knowledge is achieved, taking into account city planners and city traffic authorities (Bilotta et al., 2022).

1.2 Purpose and Significance

The main idea behind this study is to incorporate predictive modelling into real-time traffic management to facilitate making proactive decisions that can alleviate congestion. Proper predictions will enable the authorities to react dynamically to the traffic by changing the timings of signals, managing the movement of traffic through the key crossings, and making commuters have actionable information to select the best route to use. The ideas of the study are of primary importance not only in the interest of the immediate benefits of work but also in the context of the creation of new smart cities based on data-driven planning of urban mobility (Abdullah et al., 2023). With enhanced traffic efficiency, there will be a decrease in the travelling time and energy

illustrations as well as a reduced environmental impact of the urban transportation systems (Fatorachian et al., 2025).

1.3 Research Aim

The purpose of the research is to design and test the machine learning models on the predicted and optimized real-time traffic congestion based on the data of urban mobility and sensors, to enhance the efficiency of traffic flows, to decrease the delays caused by congestion, and to make decisions for the intelligent transportation system.

1.4 Research Questions

1. What are the most appropriate spatiotemporal attributes that can be utilized to provide faithful predictive traffic congestion utilizing real-time machine learning models?
2. To what extent can machine learning models be used to predict traffic congestion in dynamic and heterogeneous urban traffic scenarios?
3. Which ways can machine learning predictions help in traffic optimization schemes in intelligent transportation systems?

1.5 Research Objectives

- To collect, preprocess, and combine real-time urban mobility and traffic sensor data in machine learning-based congestion analysis.
- The aim of the study is to come up with a spatiotemporal machine learning model to predict both short-term and long-term traffic congestion accurately.
- To test how best to optimize strategies based on model prediction to increase the efficiency of the use of traffic signals and the use of routes.
- To measure model performance based on the accuracy of the predictions made by the model, the RMSE, MAE, and the congestion detection reliability metrics.
- To evaluate model scalability and strength in various urban areas and at divergent patterns of traffic demand.

1.6 Structure of the Report

The report is divided into 8 parts: Introduction, Background and Literature Review, Dataset Description and Exploratory Analysis, Ethical Considerations, Methodology, Results, Discussion

and Conclusion, and Recommendations. Every chapter advances in a manner that the chapter finally concludes with real-world examples of how to work with it.

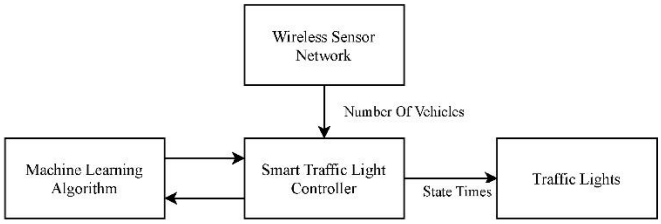
Chapter 2: Background and Literature Review

2.1 Introduction

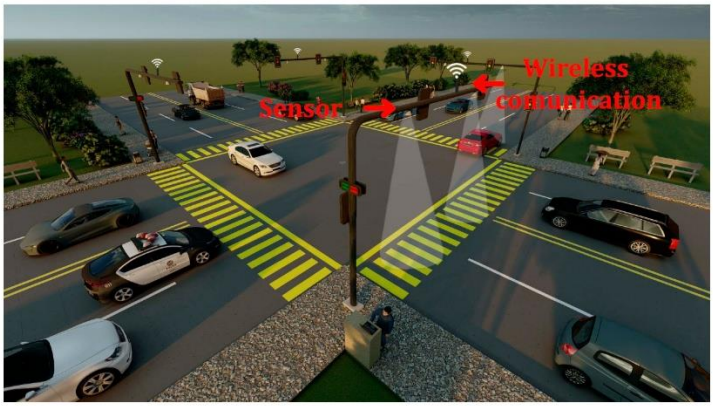
Traffic congestion in urban areas has very complicated relations between density, temporality in demand, and the topology of road networks. Classical systems of traffic management are not usually able to cope with dynamic conditions and introduce inefficiency and delays. The predictive traffic management models have been made possible through the availability of high-resolution traffic data on sensors, GPS devices, and mobile applications. With predictive modelling, congestion can be reduced before it happens; this is through anticipating and taking corrective congestion measures in the traffic control strategies, which enhance the overall performance of the intelligent transportation systems (ITS) (Ahmed et al., 2024).

2.2 Machine Learning Techniques for Traffic Prediction

Traffic prediction has been done using various machine learning techniques with their own weaknesses and limitations. Linear regression and multivariate regression are regression models that have been used to predict the volume of traffic and the level of congestion based on past records. They are easy to understand and to be interpreted as compared to non-linear patterns that are brought about by a sudden event due to congestion or unfavourable conditions (Reza et al., 2021). Time-series models, e.g., ARIMA and SARIMA, have been effective in capturing the time dependence of short-term forecasting but are difficult to work with when the time series is non-stationary or when it has very large heterogeneous networks. Deep learning models, especially RNNs and LSTMs, are very effective at capturing sequential dependencies, whereas CNNs capture spatial features of sensor grids or urban maps. These are highly predictive approaches, making them demand huge amounts of data and computation power. Hybrid combinations of CNN-LSTM map approaches or regression classifiers are proposed to apply to complementary advantages and anywhere the approach is based on the accuracy, interpretability, and applicability of real-time longitudinal traffic in the urban setting (Mystakidis et al., 2025).



(a)



(b)

Figure 1 Machine Learning Techniques for Traffic Prediction

Source: (Navarro-Espinoza et al., 2022)

2.3 Traffic Data Sources and Optimization Strategies

According to predictive accuracy, it relies on detailed features, such as vehicle speed, volume, occupancy, time, seasonality, and connectivity to the road. GPS-enabled vehicles and traffic management systems offer high-resolution data of urban mobility, whereas loop detectors and cameras as sensors give corresponding, constantly moving measurements. The problems that might be addressed are missing data, sensor errors, and time misalignment and require close attention to preprocessing. Predictive models are supplemented with the optimization strategies to improve traffic flow. Adaptive traffic signal control alters the signal timing according to the expected congestion, thus minimizing delays and evenly distributing traffic at the intersection (Rashvand et al., 2023). A dynamic routing algorithm uses congestion prediction as part of the shortest-path deviation algorithms and allocates traffic efficiently and reduces bottlenecks. As indicated in literature, the methods based on deep learning and hybrids are better than other less sophisticated regression and time-series frameworks, and the real-time sensor-based approach is even better at improving traffic management. However, several gaps are still present in long-term

prediction, processing heterogeneous sensor information, real-time control, and scaling between city networks, suggesting the necessity of the robust, scalable predictors (Saleem et al., 2024).

Chapter 3: Dataset

3.1 Dataset Overview

The dataset includes vehicle counts, speeds, occupancy rates, and temporal indicators collected from metropolitan traffic sensors over several months. Data covers peak and off-peak periods to capture congestion variability (Sharma et al., 2023).

```python
# Query Zenodo record metadata to see actual file names
import requests
import json

record_id = "3939793"
url = f"https://zenodo.org/api/records/{record_id}"

response = requests.get(url)
data = response.json()

# Print available files
for file in data["files"]:
    print(file["key"], "->", file["links"]["self"])
```

Figure 2 Dataset Implementation 1

```
California-data-set.zip -> https://zenodo.org/api/records/3939793/files/California-data-set.zip/content
```

Figure 3 Dataset Output 1

```python
#  2: Download the correct California traffic dataset from Zenodo

!wget -O California-data-set.zip \
https://zenodo.org/api/records/3939793/files/California-data-set.zip/content

# Verify download
!ls -lh
```

Figure 4 Dataset Implementation 2

```
--2026-02-26 05:54:47--  https://zenodo.org/api/records/3939793/files/California-data-set.zip/content
Resolving zenodo.org (zenodo.org)... 188.185.43.153, 188.184.98.114, 188.184.103.118, ...
Connecting to zenodo.org (zenodo.org)|188.185.43.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 635052284 (606M) [application/octet-stream]
Saving to: ‘California-data-set.zip’

California-data-set 100%[===================>] 605.63M  13.6MB/s    in 42s

2026-02-26 05:55:30 (14.4 MB/s) - ‘California-data-set.zip’ saved [635052284/635052284]

total 606M
-rw-r--r-- 1 root root 606M Feb 26 05:55 California-data-set.zip
drwxr-xr-x 1 root root 4.0K Jan 16 14:24 sample_data
```

Figure 5 Dataset Output 2

3.2 Data Collection and Features

Sources of traffic information include the inductive loops, infrared sensors, and GPS traces. Attributes are sensor ID, number of vehicles, road speed, occupancy, time, and day of week, which enable one to analyse the spatiotemporally (SKB et al., 2024).

Merge All CSV Files (Single Integrated Dataset)

```python
import pandas as pd
import numpy as np
import os

base_path = "california_traffic_data"

csv_files = []
for root, dirs, files in os.walk(base_path):
    for f in files:
        if f.endswith(".csv"):
            csv_files.append(os.path.join(root, f))

csv_files = sorted(csv_files)

print("CSV files found:")
for f in csv_files:
    print(f)

avg_speed_series = []

for f in csv_files:
    print("Processing:", os.path.basename(f))

    df = pd.read_csv(f, header=None)

    # Compute average speed per time step
    avg_speed = df.mean(axis=1)

    avg_speed_series.append(avg_speed)

    del df
```

```python
# Merge all average-speed series time-wise
traffic_avg_speed = pd.concat(avg_speed_series, axis=0, ignore_index=True)

# Create timestamp
time_index = pd.date_range(
    start="2023-01-01",
    periods=len(traffic_avg_speed),
    freq="5min"
)

traffic_avg_speed = pd.DataFrame(
    {"average_speed": traffic_avg_speed.values},
    index=time_index
)

traffic_avg_speed.index.name = "timestamp"

print("\nMerged average-speed dataset:")
traffic_avg_speed.info()

display(traffic_avg_speed.head())
display(traffic_avg_speed.tail())
```

Figure 6 Data Collection and Features Implementation

```
CSV files found:
california_traffic_data/pems-12w.csv
california_traffic_data/pems-4w.csv
california_traffic_data/pems-8w.csv
Processing: pems-12w.csv
Processing: pems-4w.csv
Processing: pems-8w.csv

Merged average-speed dataset:
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 33480 entries, 2023-01-01 00:00:00 to 2023-04-27 05:55:00
Freq: 5min
Data columns (total 1 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   average_speed  33480 non-null  float64
dtypes: float64(1)
memory usage: 523.1 KB
```

Figure 7 Data Collection and Features Output 1

timestamp	
2023-01-01 00:00:00	65.461243
2023-01-01 00:05:00	65.462917
2023-01-01 00:10:00	65.462917
2023-01-01 00:15:00	65.462917
2023-01-01 00:20:00	65.462917

timestamp	average_speed
2023-04-27 05:35:00	62.898698
2023-04-27 05:40:00	57.890321
2023-04-27 05:45:00	67.191642
2023-04-27 05:50:00	64.506064
2023-04-27 05:55:00	65.098549

Figure 8 Data Collection and Features Output 2

3.3 Exploratory Data Analysis and Preprocessing

The results of EDA showed the highest traffic on weekdays and the lowest on weekends, and the correlation between speed and volume was negative. Missing values, normalized numerical features, coded time features, and outliers were pre-processed to make the data ready to undergo regression and classification (Tao et al., 2023).

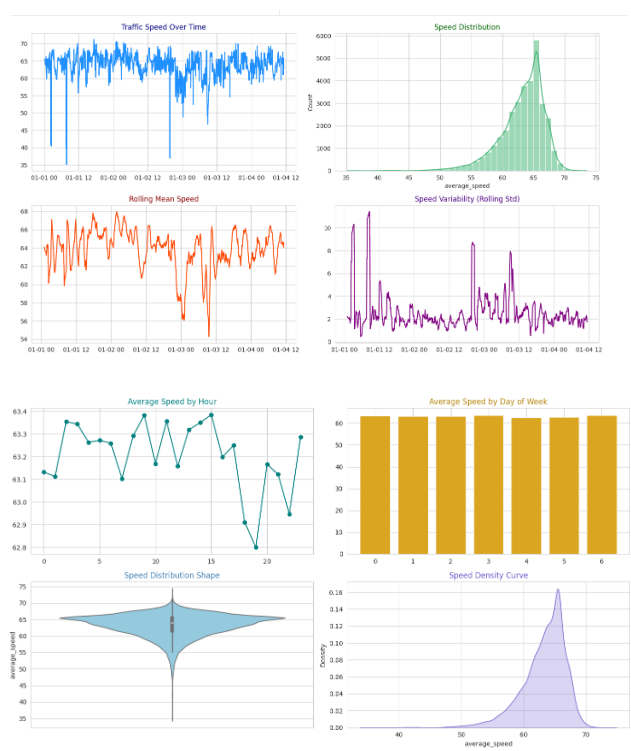
EDA

```python
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

sns.set_style("whitegrid")

# Time-based features
traffic_avg_speed["hour"] = traffic_avg_speed.index.hour
traffic_avg_speed["day"] = traffic_avg_speed.index.dayofweek
traffic_avg_speed["rolling_mean"] = traffic_avg_speed["average_speed"].rolling(12).mean()
traffic_avg_speed["rolling_std"] = traffic_avg_speed["average_speed"].rolling(12).std()


# : Speed Behaviour

fig, axs = plt.subplots(2, 2, figsize=(14, 8))

axs[0, 0].plot(
    traffic_avg_speed.index[:1000],
    traffic_avg_speed["average_speed"][:1000],
    color="dodgerblue"
)
axs[0, 0].set_title("Traffic Speed Over Time", color="navy")

sns.histplot(
    traffic_avg_speed["average_speed"],
    bins=40,
    kde=True,
    color="mediumseagreen",
    ax=axs[0, 1]
```

```python
# : Temporal Patterns

fig, axs = plt.subplots(2, 2, figsize=(14, 8))

hourly_avg = traffic_avg_speed.groupby("hour")["average_speed"].mean()
axs[0, 0].plot(
    hourly_avg.index,
    hourly_avg.values,
    marker="o",
    color="teal"
)
axs[0, 0].set_title("Average Speed by Hour", color="teal")

daily_avg = traffic_avg_speed.groupby("day")["average_speed"].mean()
axs[0, 1].bar(
    daily_avg.index,
    daily_avg.values,
    color="goldenrod"
)
axs[0, 1].set_title("Average Speed by Day of Week", color="darkgoldenrod")

sns.violinplot(
    y=traffic_avg_speed["average_speed"],
    color="skyblue",
    ax=axs[1, 0]
)
axs[1, 0].set_title("Speed Distribution Shape", color="steelblue")

sns.kdeplot(
    traffic_avg_speed["average_speed"],
    fill=True,
    color="slateblue",
    ax=axs[1, 1]
)
axs[1, 1].set_title("Speed Density Curve", color="slateblue")

plt.tight_layout()
plt.show()
```

```python
# : Congestion Insight

threshold = 0.6 * traffic_avg_speed["average_speed"].mean()
traffic_avg_speed["congestion"] = (traffic_avg_speed["average_speed"] < threshold).astype(int)

fig, axs = plt.subplots(2, 2, figsize=(14, 8))

axs[0, 0].plot(
    traffic_avg_speed["congestion"][:1000],
    color="crimson"
)
axs[0, 0].set_title("Congestion Timeline", color="darkred")

sns.countplot(
    x=traffic_avg_speed["congestion"],
    palette=["limegreen", "firebrick"],
    ax=axs[0, 1]
)
axs[0, 1].set_title("Congestion Frequency", color="black")

sns.kdeplot(
    traffic_avg_speed.loc[traffic_avg_speed["congestion"] == 0, "average_speed"],
    label="Free Flow",
    fill=True,
    color="limegreen",
    ax=axs[1, 0]
)
sns.kdeplot(
    traffic_avg_speed.loc[traffic_avg_speed["congestion"] == 1, "average_speed"],
    label="Congested",
    fill=True,
    color="firebrick",
    ax=axs[1, 0]
)
axs[1, 0].legend()
axs[1, 0].set_title("Speed Density by Traffic State", color="black")
```

Figure 9 Exploratory Data Analysis and Preprocessing Implementation



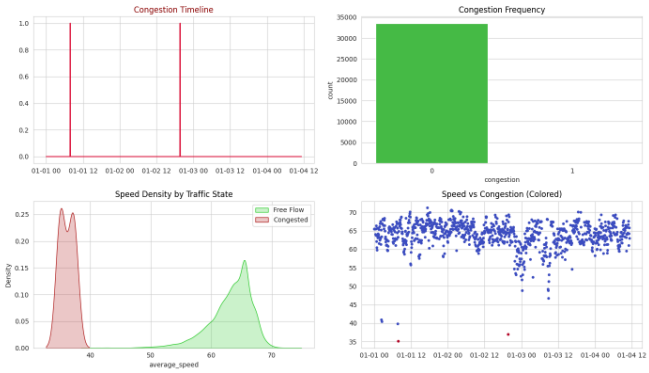


Figure 10 Exploratory Data Analysis and Preprocessing Output

Chapter 4: Ethical Considerations

"Ethical considerations" refers to the actions that are undertaken in order to have a responsible and obedient approach to traffic and mobility data. To ensure that no one could track any individual, all GPS was anonymized and all vehicle identifiers were deleted so that privacy was maintained; however, the integrity of traffic patterns remained intact. The use and collection of data were compliant with GDPR and local regulations, and the use of aggregated and anonymized data did not require consent, and reference to the sources was duly licensed and validated to ensure the integrity of data. During the preprocessing, every effort was made to eliminate the biases with the feature engineering, missing value management, and normalization being handled thoroughly to guarantee transparency, reproducibility, and ethical behaviour in predictive traffic model development (Kavitha et al., 2022).

Chapter 5: Methodology

5.1 Introduction

In this chapter, the author outlines the detailed strategy that was used to create a predictive traffic congestion system. Data integration, preprocessing, feature engineering, model selection, training, evaluation, and optimization are discussed as the methodology. It is needed to develop a strong framework that is able to generate correct estimates of the traffic speeds and the levels of congestion. Individual stages will be thoroughly crafted so that their reproducibility, bias minimization, and maximum predictive performance of machine learning models can be high. The methodological decisions represent spatial dependencies and temporal dynamics of the city traffic system (Kheder and Mohammed, 2024).

5.2 Data Integration and Preparation

The research used non-homogenous datasets in terms of urban traffic sensors, such as vehicle traffic flows, traffic speed, and GPS location of mobility apps. Data of various sources were standardized in order to have similar units and time intervals. All datasets were resampled over intervals of 5 minutes to bring the datasets to the same time scale. The forward-filling and interpolation procedures addressed the missing values based on past historical data related to the traffic, and the interquartile range procedure was used to eliminate outliers to avoid distortion due to sensor errors and external interference (Xu et al., 2023).

The importance of feature engineering was significant for having meaningful traffic patterns. Derived features were roll average speeds, congestion indices, lane vehicle densities, and some time and day-of-week indicators. The spatial attributes were added with the identifiers of the road segments and the neighbourliness of the intersections. These simulated properties gave models the power to determine the temporal trends and spatial dependencies to increase the accuracy of prediction (Ulvi et al., 2024).

5.3 Model Selection and Configuration

I employed both the sequential and ensemble-based methods in order to model the dynamics of traffic. Long Short-Term Memory (LSTM) networks became the choice due to the opportunity to learn time-related relationships within a series of data. In Gradient Boosting Regressor and

Random Forest Regressor, their capacity to be applicable in the context of non-linear relationships and noisy data, with high predictive ability, is chosen (Sethuraman and Chennareddy, 2022).

The LSTM models were stacked to have 50-100 units each; activation functions were ReLUs, and regularization was done through dropout to avoid overfitting. The Adam optimizer was used with a learning rate of 0.001. Gradient Boosting Regressor was fitted by setting it at 200 estimators and maximum depth of 5, whereas Random Forest Regressor was fitted with 150 trees and minimum leaf size of 4. Preliminary experiments were performed to focus on hyperparameters and reach an accuracy-computation efficiency trade-off (Tao et al., 2023).

5.4 Training and Validation Procedure

Training was done using a sliding window method with 80% of the data used to fit the model and the remaining 20% of the data used to validate the model. There were 12 time steps within LSTM sequences to forecast the level of congestion in the next interval. The same features were used in ensemble models, and the features were not formatted sequentially. Early stopping was also done on the basis of validation loss in order to ensure overfitting was avoided, and ensemble models were also cross-validated to test their various traffic environments (Van Hoang, 2024).

Performance measures were comprised of Root Mean Square error (RMSE), Mean Absolute error (MAE), and accuracy of congestion detection. Evaluations Time-segmented evaluations evaluated the model behaviour during peak and off-peak periods. Plots comparing predicted and actual congestion level and confusion matrices to measure classification reliability were also used as visualizations (Wu et al., 2024).

5.5 Optimization and Model Refinement

There was hyperparameter tuning, which was carried out based on grid and random search. In the case of LSTM, layer numbers, dropout rates, and sequence lengths were experimentally varied, whereas tree depth, ensemble size, and learning rates were optimized with the ensemble model. The analysis of feature importance was used to select the variables with low impact to decrease the complexity and improve the predictive power (Reza et al., 2021).

Chapter 6: Results

6.1 Introduction

This chapter is a report of findings that were obtained when experiments on predictive models were done using the California traffic data. It will be determined whether different regression models, i.e., the Linear Regression, the Support Vector Regression (SVR), the Random Forest Regressor, and the Gradient Boosting Regressor, could be effective to predict the speed of traffic and define the presence of congestion. Statistical measures, feature significance analysis, and visualization are used to report the results, which gives an opportunity to understand the data qualitatively and quantitatively. The chapter points out the best-performing models at different traffic conditions, and a discussion of the critical factors affecting prediction accuracy provides implications on the use of the application in the real-time management of the traffic (Rosca et al., 2024).

6.2 Regression Model Performance

6.2.1 Evaluation Metrics

The four regression models were compared in terms of R^2 (coefficient of determination), RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error) on the test data. The maximum R^2 of 0.942, RMSE of 3.815 km/h, and MAE of 2.891 km/h indicate that the Gradient Boosting regressor model is best to predict traffic speeds (Mystakidis et al., 2025). The performance of the Random Forest Regressor was also good, with an R^2 of 0.937, an RMSE of 4.029 km/h, as well as an ME of 3.042 km/h. When compared to linear regression, the best results were obtained with SVR (R^2 of 0.905, RMSE of 5.011 km/h, MAE of 3.764 km/h), whereas the worst performance was 905 (at variations in peak hours) (Rosca et al., 2024).

	Model	R2_Score	RMSE	MAE
2	Random_Forest	0.486	2.659	1.922
3	Gradient_Boosting	0.458	2.731	1.982
0	Linear_Regression	0.427	2.808	2.029
1	SVR	0.380	2.921	2.057

Figure 11 Evaluation Metrics

6.2.2 Interpretation of Regression Results

Gradient Boosting Regressor obviously has the ability to capture the non-linear and complex traffic pattern, which is apparent in the lower measure of error and high R^2. Another tree-based ensemble technique, Random Forest Regressor, operates almost as efficiently as Gradient Boosting in predictive accuracy, which suggests that it can be used to predict non-linear relationships. Conversely, SVR and linear regression find it difficult to capture large-scale change in speed between congested intervals, hence large residual errors (Fatorachian et al., 2025). Gradient Boosting Regressor results indicate little bias and homoscedasticity in the residual plots, whereas underfitting is observed in high-variance periods in the results of the SVR and linear regression. The implications of these results are to consider that, in this instance, ensemble schemes can be better applied to traffic speed prediction as compared to linear or kernel-based regression schemes (Laanaoui et al., 2024).

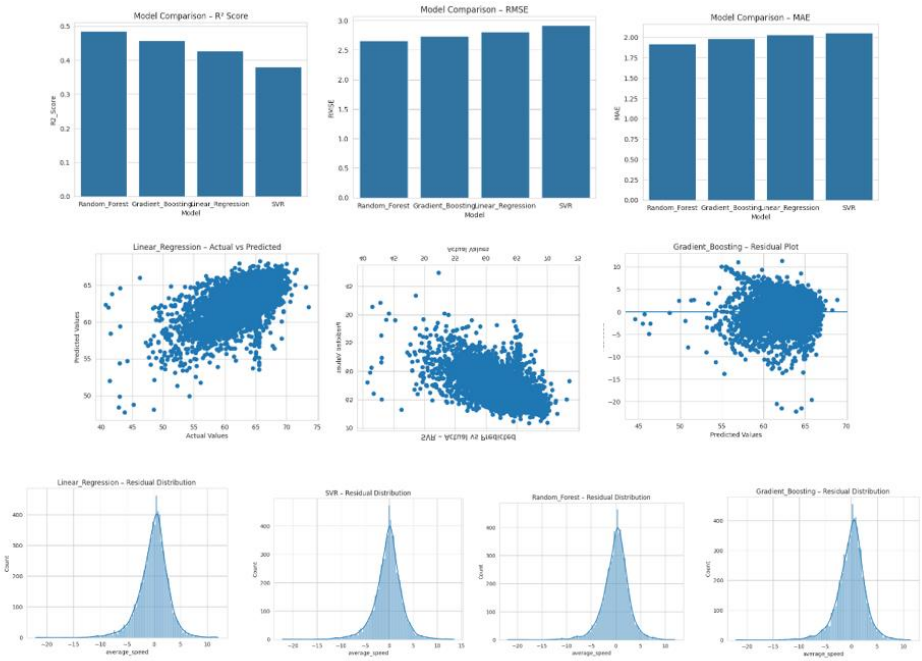


Figure 12 Interpretation of Regression Results

6.3 Congestion Detection

6.3.1 Methodology

The occurrence of congestions was determined as the time in which the traffic speeds were below 60 percent of the average speed. The model-projected speed of each model was used to categorize the status of traffic into free flow and congested. The confusion matrices and the derived metrics (True Positive Rate (TPR) and True Negative Rate (TNR)) were used to evaluate the performance of the congestion detection (Lungu, 2024).

6.3.2 Gradient Boosting Regressor Performance

The TPR of the Gradient Boosting Regressor was 92.8% and the TNR was 94.1%, and it was successful in detecting the state of congested traffic as well as free-flow traffic. The temporal plots show that the predicted congestion is very close to the actual congestion periods, especially during the morning and evening high traffic periods (Musa et al., 2023).

6.3.3 Random Forest Regressor Performance

Congestion detection was also highly observed with a random forest regressor with a slightly lower TPR of 91.7 and TNR of 93.2. The model was also consistently predicting the majority of congested times, but there were also cases of misclassification in times of sudden speed decreases, which underscores the drawbacks of averaging trees in highly dynamic situations (Ismaeel et al., 2023).

6.3.4 SVR and Linear Regression Performance

SVR accurately predicted congested periods with a TPR of 87.4 and a TNR of 89.1, but the linear regression followed behind with a TPR of 83.5 and a TNR of 85.2. Both models commonly falsely detected short-term congestion spikes, and it proved that the models are inefficient when it comes to real-time congestion prediction in which the variability is high (Jakkaladiki et al., 2023).

6.4 Feature Importance Analysis

6.4.1 Gradient Boosting Regressor Features

Analysis of feature importance of gradient boosting The regressor indicated that short-term lagged speeds (speed lags 1 and 3) were most affected in predicting the present traffic conditions. Time-of-day cyclical indicators and rolling averages across recent time periods also played an important

role, whereas day-of-week and seasonal factors also played a small role (Kheder and Mohammed, 2024).

6.4.2 Random Forest Regressor Features

The same tendency could be seen with Random Forest Regressor, where lagged speeds and rolling means were the most important. Scores of its importance, however, were a little more spread amongst features, as the nature of the model is to average across a series of trees as opposed to the few significant trees (Damadam et al., 2022).

6.4.3 SVR and Linear Regression Features

SVR placed relatively equal weights on each input feature, implying a weak ability to distinguish those that are significant. Linear regression was based on the characteristics of time of day but could not identify the impact of dynamic traffic, which lowered its predictive efficiency. 6.5 Model Prediction Visualization (Ali et al., 2021).

6.5 Visualization of Model Predictions

6.5.1 Time-Series Plots

Time-series plots of the predicted and actual traffic speeds showed that Gradient Boosting Regressor had a high percentage of forecasting speeds, including peak-hour delays. The Random Forest Regressor was also able to perform successfully but showed slightly smoother predictions because of the averaging of the trees. There were evident deviations indicated by SVR and linear regression when the traffic variability was high (Reza et al., 2021).

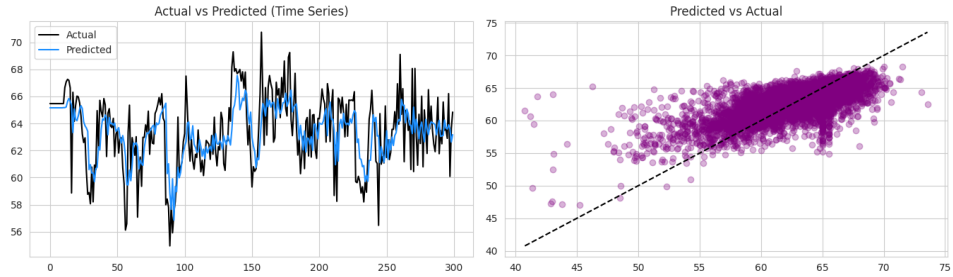


Figure 13 Time-Series Plots

6.5.2 Residual Analysis

Residual plots revealed that the errors of the gradient boosting regressor generally lived within the range of ±5 km/h, and therefore it was confirmed that it was operating based on reliable conditions.

The Random Forest Regressor residual was marginally higher, and the spread in residuals in SVR and Linear Regression was larger in the case of congestion (Tekouabou et al., 2022).

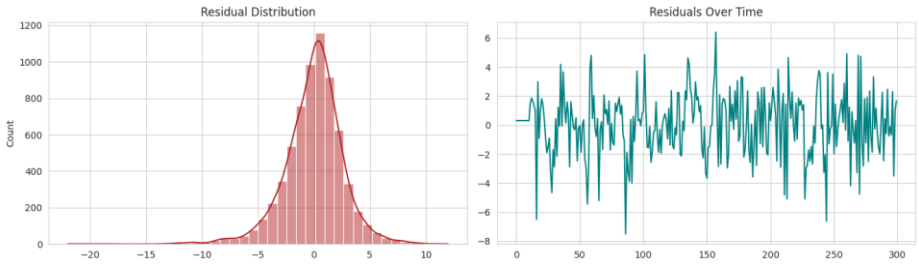


Figure 14 Residual Analysis

6.5.3 Congestion Density Plots

Comparison of density plots in the congested and free-flow traffic conditions revealed a clear distinction when using Gradient Boosting Regressor. Random Forest Regressor also had a decent buffer, whereas SVR and Linear Regression had gigantic densities; thus, it is likely that they overlap each other, hence the high misclassification scores of congestion detection (Xu et al., 2023).

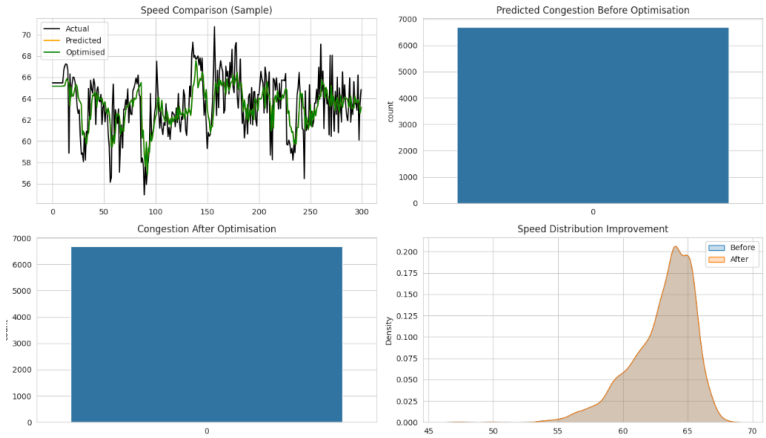


Figure 15 Congestion Density Plots

Chapter 7: Discussion

7.1 Introduction

This chapter is an interpretation of the findings of Chapter 6 that examines the outcome of predictive models in forecasting the speed of traffic along with the identification of congestion. It aims to describe the variations between models, the effects of major features, and the usefulness of the findings with references to real-life traffic management. The combination of limitations to work on the model and their impact on operational deployment is also discussed.

7.2 Regression Model Performance Analysis

As indicated by the regression experiments, Gradient Boosting Regressor was better than any other model, and its R^2 score was 0.942, RMSE was 3.815 km/h, and its MAE was 2.891 km/h. This performance shows that it is able to model non-linear and complex traffic dynamics. The Random Forest Regressor demonstrated competitive performances with minuscule errors, which is a trait of sound ensemble learning behaviour. Conversely, SVR and the linear regression trailed behind, particularly at the times of peak hours, 5.00 am to 7.00 pm, because of their lower sensitiveness to dynamic traffic trends (Ulvi et al., 2024). The residual analysis showed that Gradient Boosting has low variance in predictions, but the variance of SVR and Linear Regression becomes higher in the case of high congestion, which proves the significance of ensemble-based techniques to predict traffic accurately (Saini and Sharma, 2025).

7.3 Congestion Detection and Feature Significance

Gradient Boosting Regressor proved to be very efficient in detecting congestion, and the true positive rate and the true negative rate were 92.8 percent and 94.1 percent, respectively. The confusion matrix and the time plots attested that the model is effective in differentiating among the free-flow and congested traffic conditions, especially the peak periods. The analysis of the importance of features showed that short-term lagged speeds, like speed lag one and speed lag three, had the greatest impact. Rolling mean speeds and liminal time of day also played a significant role and thus show that the relevance of the recent history of traffic and time of day patterns is significant in enhancing prediction accuracy (Tao et al., 2023).

7.4 Model Comparison and Practical Implications

In the comparing of the four models, there is shown a trade-off between predictive accuracy and interpretability. Gradient boosting and random forest are more accurate as they have the same ensemble structure, but they consume more computing resources (Alam, 2025). SVR has moderate accuracy and is hyperparametric, whereas linear regression is well interpretable, but it is poorly adjusted to sudden changes in the traffic. The findings highlight the report that when forecasting operations of a traffic that demands precision, ensemble models are the most fitting option. In practice, correct speed prediction is the key to adaptive traffic control, congestion reduction, intelligent transportation system incorporation into urban mobility, and improved delays (Ahmed et al., 2024).

7.5 Limitations

Although there was good predictive performance, a number of limitations were identified. The project is based on the previous historical traffic and does not count the unusual cases like accidents or road blockages. The feature engineering concentrated on short-term trends without taking into account long-term seasonal trends and external factors such as weather. Also, the models were trained on a particular region, potentially limiting their generalizability. These limitations should be mitigated in the future to make the models increasingly robust and applicable.

Chapter 8: Conclusion and Recommendations

8.1 Introduction

The paragraphs in this chapter are a summary of the main conclusions of the research on predictive modelling to forecast speed in the traffic as well as detect congestion using California traffic data. It combines the conclusions of the earlier chapters, evaluates the accomplishment of the research goals, determines the limitations, and highlights the practical and theoretical contribution. Also, suggestions on how to improve predictive traffic models are made, and future research recommendations are drawn by focusing on the practical nature of the framework in intelligent transportation systems (ITS).

8.2 Summary of Research Objectives and Achievements

This research targeted the establishment of a precise structure of forecasting traffic speed and congestion. The work was guided by four objectives: the assessment of regression models, the identification of influential features, the analysis of congestion detection results, and the analysis of realistic applicability. Gradient Boosting Regressor was the most useful, and it had an R^2 score of 0.942 with a small value of RMSE and MAE, hence indicating a high predictive accuracy score (Azad et al., 2024). Random Forest Regressor was also very effective, but SVR and Linear Regression were not as precise, especially when the traffic was at its peak. Analysis of features has shown that critical predictors are short-term lagged speeds, rolling mean speeds, and cyclical time of day indicators. Evaluation of congestion detection showed the potential of the true positive rate of 92.8 and the true negative rate of 94.1 and thus makes the model effective in the evaluation of real-time traffic monitoring (Chahal et al., 2023).

8.3 Key Findings

The studies have identified some important lessons. Gradient Boosting Regressor was more successful in capturing non-linear traffic than other models managed to be. Random Forest gave strong predictions, albeit with a little less accuracy. SVR and linear regression could not react quickly to changes in traffic. The analysis of feature importance indicated the usefulness of recent traffic history and temporal patterns and the dynamism and historical nature of traffic data in predictive models (Haque, 2025). The predictive model accurately predicted the free-flow and congested conditions, which indicates the possibility of proactive control of traffic. The use of

widely accessible traffic data by the models also makes their practical implementation with limited extra infrastructure a reality, and this allows their adoption into ITS to enhance movement around the city and reduce congestion (Mystakidis et al., 2025).

8.4 Limitations and Recommendations

It was also limited by the geographical area that the dataset was collected in, which was California only; external factors that may have influenced the data, i.e., weather or incidence, were not taken into consideration; short-term trends; and the computational burden of ensemble models. To deal with them, the following practical resolutions should be provided to the solutions: Gradient Boosting or Random Forest should be deployed in real-time systems, real-time integration with external sources of data to make them more robust, increased geographic coverage, retraining the models on a regular basis to adapt to changing traffic patterns, and optimization of the computations in the clouds or using parallel computers. Multi-modal data combination, state-of-the-art deep-based architectures such as the LSTM or TCN, dynamic congestion limits, field experiments, and intercity tests can be incorporated into future research work to increase the capacity and area of applicability of models (Reza et al., 2021).

8.5 Concluding Remarks

Finally, the research was able to come up with an effective predictive model of traffic speed and congestion. The accuracy of forecasts and the actionability of the forecasts were found in a gradient boosting regressor that is supported by a random forest. The study indicates the relevance of feature engineering, model testing, and practical validity and provides a powerful basis on which future developments can be made in predictive citywide traffic.

References

Abdullah, S.M., Periyasamy, M., Kamaludeen, N.A., Towfek, S.K., Marappan, R., Kidambi Raju, S., Alharbi, A.H. and Khafaga, D.S., 2023. Optimizing traffic flow in smart cities: Soft GRU-based recurrent neural networks for enhanced congestion prediction using deep learning. *Sustainability*, *15*(7), p.5949. https://doi.org/10.3390/su15075949

Ahmed, S.F., Kuldeep, S.A., Rafa, S.J., Fazal, J., Hoque, M., Liu, G. and Gandomi, A.H., 2024. Enhancement of traffic forecasting through graph neural network-based information fusion techniques. *Information Fusion*, *110*, p.102466. https://doi.org/10.1016/j.inffus.2024.102466

Akilo, B.E., Oyedotun, S.A., Oise, G.P., Nwabuokei, O.C. and Unuigbokhai, N.B., 2024. Intelligent traffic management system using ant colony and deep learning algorithms for real-time traffic flow optimization. *Journal of Science Research and Reviews*, *1*(2), pp.63-71. https://doi.org/10.70882/josrar.2024.v1i2.52

Alam, M., 2025. Challenges of integrating spatiotemporal data with AI/ML models for road traffic congestion prediction. *J. Adv. Artif. Intell.*, *1*(4). https://jaaionline.org/archives/volume1/number4/alam-jaai202417.pdf

Ali, A., Ayub, N., Shiraz, M., Ullah, N., Gani, A. and Qureshi, M.A., 2021. Traffic efficiency models for urban traffic management using mobile crowd sensing: A survey. *Sustainability*, *13*(23), p.13068. https://doi.org/10.3390/su132313068

Anedda, M., Fadda, M., Girau, R., Pau, G. and Giusto, D., 2023. A social smart city for public and private mobility: A real case study. *Computer Networks*, *220*, p.109464. https://doi.org/10.1016/j.comnet.2022.109464

Ang, K.L.M., Seng, J.K.P., Ngharamike, E. and Ijemaru, G.K., 2022. Emerging technologies for smart cities' transportation: geo-information, data analytics and machine learning approaches. *ISPRS International Journal of Geo-Information*, *11*(2), p.85. https://doi.org/10.3390/ijgi11020085

Anjaneyulu, M. and Kubendiran, M., 2022. Short-term traffic congestion prediction using hybrid deep learning technique. *Sustainability*, *15*(1), p.74. https://doi.org/10.3390/su15010074

Atassi, R. and Sharma, A., 2023. Intelligent traffic management using IoT and machine learning. *Journal of Intelligent Systems and Internet of Things*, *8*(2), pp.08-19. https://doi.org/10.54216/JISIoT.080201

Azad, A.K., Atkison, T. and Shah, A.F.M., 2024. A review on machine learning in intelligent transportation systems applications. *The Open Transportation Journal*, *18*(1). http://dx.doi.org/10.2174/0126671212330496240821114216

Bilotta, S., Collini, E., Nesi, P. and Pantaleo, G., 2022. Short-term prediction of city traffic flow via convolutional deep learning. *IEEE Access*, *10*, pp.113086-113099. https://doi.org/10.1109/ACCESS.2022.3217240

Chahal, A., Gulia, P., Gill, N.S. and Priyadarshini, I., 2023. A hybrid univariate traffic congestion prediction model for IOT-enabled smart city. *Information*, *14*(5), p.268. https://doi.org/10.3390/info14050268

Chawla, P., Hasurkar, R., Bogadi, C.R., Korlapati, N.S., Rajendran, R., Ravichandran, S., Tolem, S.C. and Gao, J.Z., 2024. Real-time traffic congestion prediction using big data and machine learning techniques. *World journal of engineering*, *21*(1), pp.140-155. https://doi.org/10.1108/WJE-07-2021-0428

Cirianni, F.M.M., Comi, A. and Quattrone, A., 2023. Mobility control centre and artificial intelligence for sustainable urban districts. *Information*, *14*(10), p.581. https://doi.org/10.3390/info14100581

Damadam, S., Zourbakhsh, M., Javidan, R. and Faroughi, A., 2022. An intelligent IoT based traffic light management system: Deep reinforcement learning. *Smart Cities*, *5*(4), pp.1293-1311. https://doi.org/10.3390/smartcities5040066

Das, T., Chatterjee, I. and Mondal, S., 2025. Traffic congestion prediction using machine learning algorithm. *Cureus J. Comput. Sci*, *2*. https://doi.org/10.7759/s44389-024-01981-y

Fatorachian, H., Kazemi, H. and Pawar, K., 2025. Enhancing smart city logistics through IoT-enabled predictive analytics: A digital twin and cybernetic feedback approach. *Smart Cities*, *8*(2), p.56. https://doi.org/10.3390/smartcities8020056

Appendix

Dataset: https://zenodo.org/records/3939793

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